

A CERTIFIED COMPUTATIONAL VERIFICATION FOR ERDŐS PROBLEM #458

AUTHOR NAME (FILL IN)

ABSTRACT. Erdős and Graham asked whether $\text{lcm}(1, \dots, p_{k+1} - 1) < p_k \text{lcm}(1, \dots, p_k)$ for every k , where p_k is the k -th prime. We verify this for all k with $p_{k+1} \leq 10^{21}$. The proof is self-contained: it locates every prime power with exponent at least 3 below the bound exactly and exhibits a Pocklington-certified prime between every pair of consecutive prime squares, so it does not rely on tables of maximal prime gaps. Every claim is backed by a certificate that was re-validated by an independent implementation.

1. THE PROBLEM

Erdős Problem #458 [1] asks the following.

Let $[1, \dots, n]$ denote the least common multiple of $\{1, \dots, n\}$ and p_k the k -th prime. Is it true that, for all $k \geq 1$,

$$[1, \dots, p_{k+1} - 1] < p_k [1, \dots, p_k]?$$

At the time of writing it is listed as *falsifiable* in the community database [2]: it is open, but a counterexample, if one exists, is a finite object.

Prior work. The forum thread for the problem contains 3 comments. The largest verification bound reported there is 10^{20} (bhowerton, erdosproblems.com forum, 12 Jun 2026; that verification relies on prime-gap records above $4e18$). The present computation goes beyond that bound.

2. METHOD

Reduction. For consecutive primes $p < r$ the primes dividing $[1, \dots, r - 1]/[1, \dots, p]$ are exactly the bases q of prime powers q^a ($a \geq 2$) in the open interval (p, r) , each contributing one factor q per such power. Hence the inequality for the gap (p_k, p_{k+1}) is equivalent to $\Pi(p_k, p_{k+1}) < p_k$, where $\Pi(p, r) = \prod_{q^a \in (p, r)} q$. (This reformulation also appears in the forum discussion of the problem.)

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Certificate. We prove the statement for all k with $p_{k+1} \leq X$ without using any table of maximal prime gaps:

- (A) For every prime power $h = q^a \leq X$ with $a \geq 3$ we determine the enclosing gap $(p, r) \ni h$ exactly (deterministic primality testing of every integer between), list *all* prime powers inside it, squares included, and verify $\Pi(p, r) < p$.
- (B) For every pair of consecutive primes $q < q'$ with $q \leq \sqrt{X}$ we exhibit a prime in (q^2, q'^2) . Consequently no prime gap contains two prime squares.
- (C) A gap with $r \leq X$ containing no prime power of exponent ≥ 3 thus contains at most one prime square q^2 , so $\Pi(p, r) \leq q < p$ (there is a prime in (q, q^2) by Bertrand's postulate).

Primality. All primality decisions use either the Miller–Rabin test with the first 13 prime bases, which is deterministic below $3.317 \cdot 10^{24}$ [7], or a Pocklington–Lehmer certificate [6, 3]: the witnesses in (B) are chosen of the form $N = Pm + 1$ with P one of the 32 largest primes $\leq q$ and m even, so that $F = 2P > \sqrt{N}$ is a completely factored divisor of $N - 1$ and a single base a with $a^{N-1} \equiv 1$, $\gcd(a^{(N-1)/P} - 1, N) = \gcd(a^{(N-1)/2} - 1, N) = 1$ proves N prime. Compositeness is only ever asserted from an explicit failed strong-probable-prime test, which is a proof.

Independent checking. The search is implemented in C (two-limb Montgomery arithmetic). A separate pure-Python implementation, sharing no code with it, (i) re-verifies the literal lcm inequality and the reduction on an initial range, (ii) recomputes part (A) in full and compares a SHA-256 digest of all gap records, and (iii) recomputes randomly sampled chunks of part (B), comparing a 64-bit FNV-1a digest of the entire witness sequence. The witness rule is a pure function of (q, q') , so agreement of digests means both implementations produced identical witnesses.

3. RESULTS

Theorem 1. *For every k with $p_{k+1} \leq 10^{21}$,*

$$[1, \dots, p_{k+1} - 1] < p_k [1, \dots, p_k].$$

Part (A) located 683532 prime powers with exponent ≥ 3 , lying in 683531 distinct prime gaps, of which 5 contain more than one prime power. Part (B) exhibited 1367199811 witnesses (one per prime $q \leq 31622776601$), in 3163 chunks; 3782301 of them needed the Miller–Rabin fallback rather than a Pocklington certificate. The square scan used about 6.2 CPU-hours. The literal lcm inequality was additionally checked directly for $p_{k+1} \leq 3000$. Limitation: the independent re-implementation recomputed part (A) in full but part (B) only for a random sample of chunks together with the first and last (Section 4); the coverage of (B) is confirmed by comparing the number of witnesses with $\pi(\lfloor \sqrt{X} \rfloor)$ computed by a sieve-free method, while correctness

of the witnesses outside the sample rests on the C implementation and its per-witness primality proofs.

The gaps with the largest ratio $\Pi(p, r)/p$ are all small:

gap (p, r)	prime powers inside	$\Pi(p, r)$
(7, 11)	$2^3 \cdot 3^2$	6
(23, 29)	$3^3 \cdot 5^2$	15
(113, 127)	$5^3 \cdot 11^2$	55
(13, 17)	2^4	2
(31, 37)	2^5	2
(79, 83)	3^4	3

4. VERIFICATION AND REPRODUCIBILITY

The certificate produced by the search was re-validated by an independent implementation (“p458-independent-python”) that shares no code with the search:

- literal lcm inequality re-verified for $p_{(k+1)} \leq 3000$
- reduction (product criterion) agrees with literal lcm on that range
- (A) 683532 prime powers / 683531 gaps recomputed independently; digest matches
- (B) 3163 chunks tile $[2, 31622776601]$ with 1367199811 primes, no failures reported
- (B) witness count equals $\pi(31622776601) = 1367199811$, computed by the Lucy_Hedgehog recursion (no sieve): every prime $q \leq \sqrt{X}$ has a witness
- 26 chunks (including the first and the last, the others chosen at random) recomputed from scratch and reproduced exactly: 11,182,376 witnesses in total

The raw certificate (`result.json`, SHA-256 08344b04871cb70f cc2b50e4a69dc9b2 a7f39ac6a4eb6449 b7adf266b88258cc) and the check report are deposited with the software.

USE OF AI TOOLS

The search and checking software was written with the assistance of an AI coding assistant (Claude, Anthropic). All numerical claims come from executed runs whose certificates were re-validated by an independent implementation; the author(s) reviewed the code and the mathematics and take full responsibility for the content.

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AFFILIATION (FILL IN)

Email address: email@example.org